Vectors

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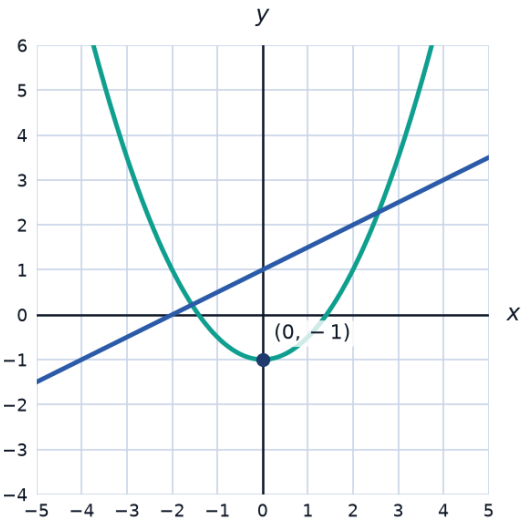
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$ and $|\mathbf{a}|$.

2 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.



3 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find $2\mathbf{a} - \mathbf{b}$ for $\mathbf{a} = (1, 3, -2)$ and $\mathbf{b} = (4, -1, 2)$.

4 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

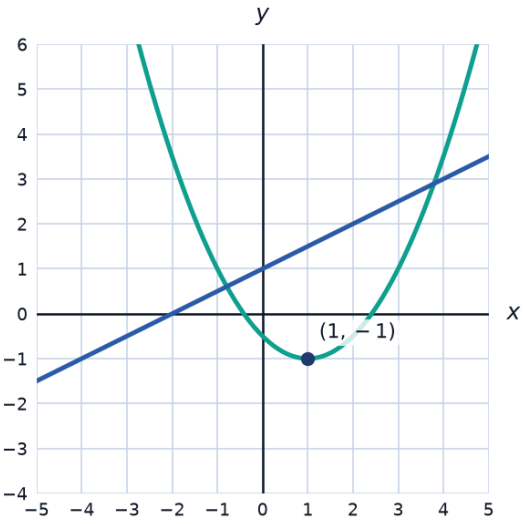
5 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.

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6 [Foundational] • Target: The scalar (dot) product and the angle between vectors

Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.



7 [Foundational] • Target: The scalar (dot) product and the angle between vectors

Find the angle between $\mathbf{a} = (3, 0, 4)$ and $\mathbf{b} = (0, 5, 0)$, to the nearest degree.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The vector equation of a line

Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.

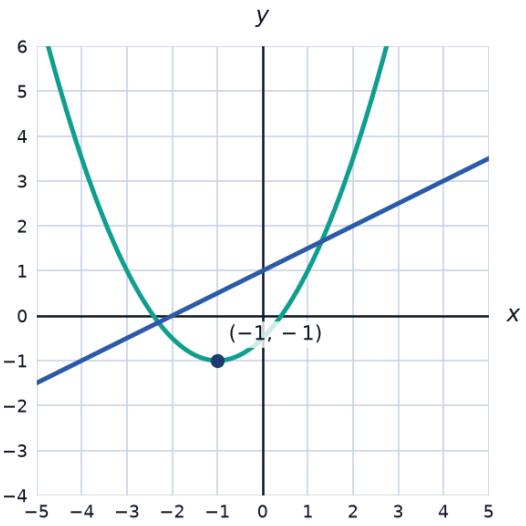
9 [Intermediate] • Target: The vector equation of a line

Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.

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10 [Intermediate] • Target: The vector equation of a line

Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.



11 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.

12 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.

13 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.

14 [Intermediate] • Target: Intersecting and skew lines

Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

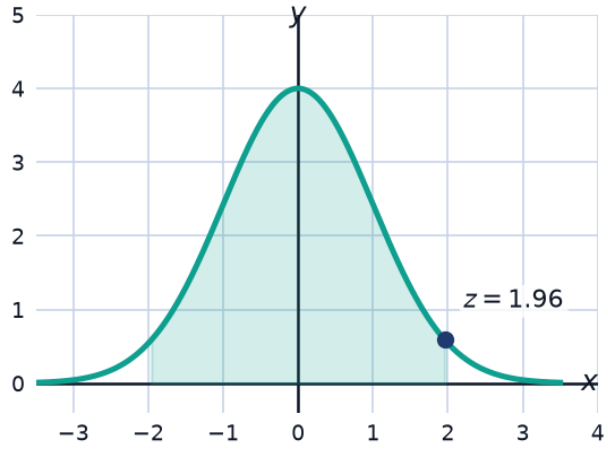
15 [Challenging] • Target: Intersecting and skew lines • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.

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16 [Challenging] • Target: The equation of a plane • Cross-Lesson: 9.01 Complex Numbers

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.



17 [Challenging] • Target: The equation of a plane • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.

18 [Challenging] • Target: The equation of a plane • Cross-Lesson: 10.01 Numerical Methods

Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.

19 [Challenging] • Target: The equation of a plane • Cross-Lesson: 11.01 Differential Equations

Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.

20 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 12.01 Poisson Distribution

Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

Given $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$, find the unit vector in the direction of $\mathbf{a}$.

a. $\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$

b. $\frac{2}{49}\mathbf{i} - \frac{3}{49}\mathbf{j} + \frac{6}{49}\mathbf{k}$

c. $\frac{1}{7}(2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})$

d. $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

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22 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

Find the distance between the points $P(1, -2, 4)$ and $Q(4, 2, -8)$.

a. 13

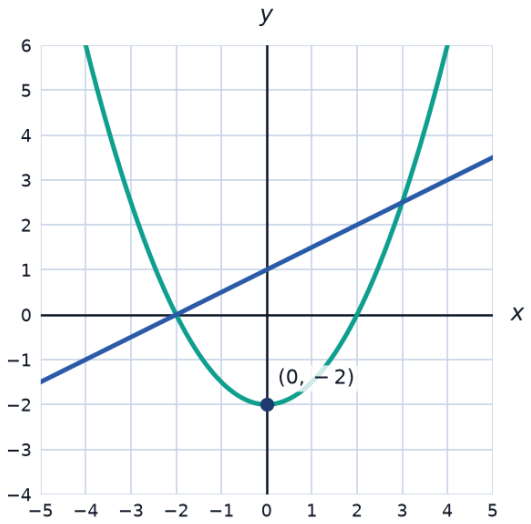
b. 12

c. $\sqrt{153}$

d. 14

23 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

If $\mathbf{u} = [1, 2, -1]$ and $\mathbf{v} = [3, -1, 2]$, compute $2\mathbf{u} + 3\mathbf{v}$.



a. $[11, 1, 4]$

b. $[11, 7, 4]$

c. $[8, 1, 4]$

d. $[7, 3, 2]$

24 [Foundational] • Target: The Scalar (Dot) Product & Angles

Compute the scalar product $\mathbf{a} \cdot \mathbf{b}$ where $\mathbf{a} = [3, -2, 4]$ and $\mathbf{b} = [1, 5, 2]$.

a. 1

b. 5

c. -1

d. 15

25 [Foundational] • Target: The Scalar (Dot) Product & Angles

For what value of k are the vectors $\mathbf{u} = [2, k, -3]$ and $\mathbf{v} = [4, 2, 6]$ perpendicular?

a. $k = 5$

b. $k = -5$

c. $k = 10$

d. $k = 0$

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26 [Foundational] • Target: The Scalar (Dot) Product & Angles

Find the acute angle between vectors $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [0, 1, 1]$.

a. 60°

b. 45°

c. 30°

d. 90°

27 [Foundational] • Target: Vector Equations of Lines in 3D

Find the vector equation of the line passing through point $A(1, 2, 3)$ in the direction $\mathbf{d} = [2, -1, 4]$.

a. $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$

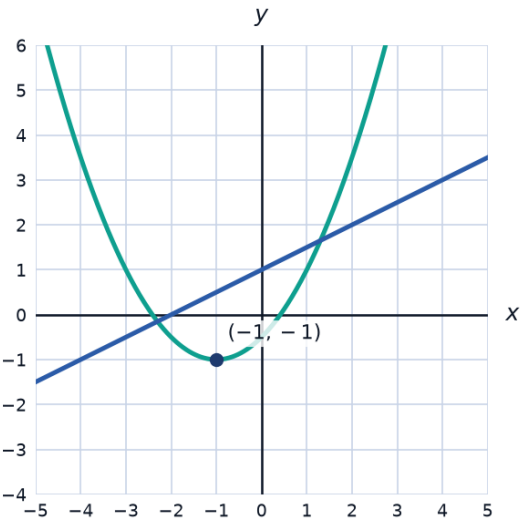
b. $\mathbf{r} = [2, -1, 4] + \lambda[1, 2, 3]$

c. $\mathbf{r} = [1, 2, 3] \cdot [2, -1, 4]$

d. $\mathbf{r} = [3, 1, 7] + \lambda[2, -1, 4]$

28 [Foundational] • Target: Vector Equations of Lines in 3D

Determine whether the point $(5, 0, 11)$ lies on the line $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$.



a. Yes, corresponding to $\lambda = 2$

b. No, λ values are inconsistent

c. Yes, corresponding to $\lambda = 3$

d. No, because the z-coordinate is too large

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Vector Equations of Lines in 3D**

Find the point of intersection of the lines $\mathbf{r}_1 = [1, 0, 1] + \lambda[1, 2, -1]$ and $\mathbf{r}_2 = [0, 4, 3] + \mu[2, -1, -2]$.

a. $(2, 2, 0)$

b. $(1, 0, 1)$

c. $(3, 4, -1)$

d. The lines are skew

30 **[Intermediate]** • **Target: Skew Lines & Shortest Distances**

Two lines in 3D do not intersect and are not parallel. What term describes such lines?

a. Skew lines

b. Coplanar lines

c. Asymptotic lines

d. Coincident lines

31 **[Intermediate]** • **Target: Skew Lines & Shortest Distances**

Find the acute angle between the line $\mathbf{r} = [1, 0, 2] + \lambda[1, -2, 2]$ and the line $\mathbf{r}' = [3, -1, 4] + \mu[3, 0, 4]$.

a. $\arccos(11/15) \approx 42.8^\circ$

b. 30°

c. 45°

d. 60°

32 **[Intermediate]** • **Target: Skew Lines & Shortest Distances**

Find the coordinates of the foot of the perpendicular from the origin to the line $\mathbf{r} = [1, -1, 2] + \lambda[2, 1, -1]$.

a. $[1/3, -4/3, 7/3]$

b. $[1, -1, 2]$

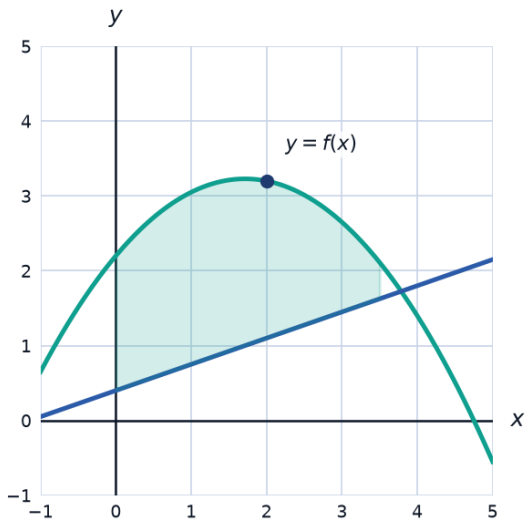
c. $[0, 0, 0]$

d. $[2, 1, -1]$

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33 [Intermediate] • Target: Skew Lines & Shortest Distances

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2)$, $L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?



a. $3\sqrt{5}$

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

34 [Intermediate] • Target: Vector arithmetic in three dimensions

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2

b. 2

c. -4

d. 4

35 [Intermediate] • Target: The scalar (dot) product and the angle between vectors

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k-y}{2x+1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1

b. 11

c. $\frac{5}{4}$

d. 5

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36 [Intermediate] • Target: The vector equation of a line

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

37 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

a. 7

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Intersecting and skew lines • Cross-Lesson: 10.01 Numerical Methods

Two lines have direction vectors $\mathbf{d_1} = (1, -3, -3)$ and $\mathbf{d_2} = (4, -3, 1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. parallel

b. identical

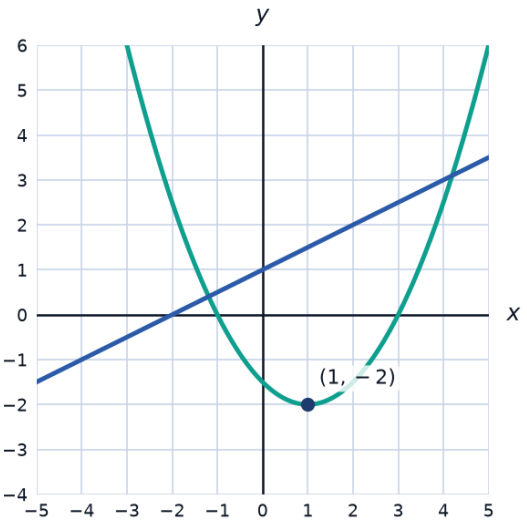
c. perpendicular

d. neither parallel nor perpendicular

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39 [Challenging] • Target: The equation of a plane • Cross-Lesson: 11.01 Differential Equations

Two lines have direction vectors $\mathbf{d_1} = (-4, -3, 4)$ and $\mathbf{d_2} = (-4, -5, -5)$. What is the relationship between the two lines (assuming they are coplanar)?



a. perpendicular

b. neither parallel nor perpendicular

c. identical

d. parallel

40 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 12.01 Poisson Distribution

Two lines have direction vectors $\mathbf{d_1} = (-7, -3, -1)$ and $\mathbf{d_2} = (7, -3, 2)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular

b. identical

c. perpendicular

d. parallel

41 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 13.01 Continuous Random Variables

Two lines have direction vectors $\mathbf{d_1} = (1, -2, 2)$ and $\mathbf{d_2} = (-5, 6, 6)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. neither parallel nor perpendicular

c. parallel

d. perpendicular

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42 [Challenging] • Target: A well-designed word problem: a drone's flight path • Cross-Lesson: 14.01 Sampling & Estimation

Two lines have direction vectors $\mathbf{d_1} = (1, -2, 1)$ and $\mathbf{d_2} = (3, 3, -1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. parallel

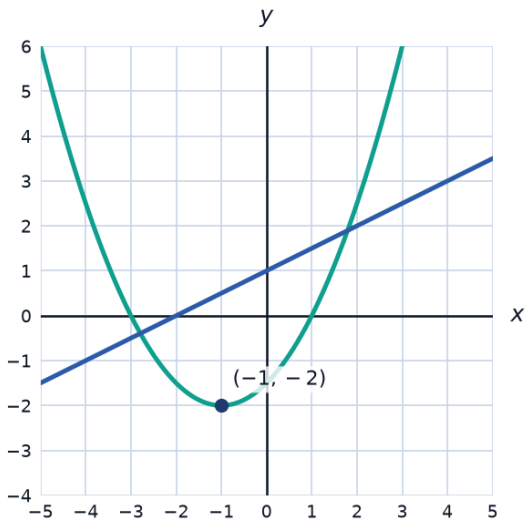
b. neither parallel nor perpendicular

c. identical

d. perpendicular

43 [Challenging] • Target: 3D Vector Arithmetic & Magnitude • Cross-Lesson: 15.01 Hypothesis Testing

Two lines have direction vectors $\mathbf{d_1} = (6, 2, 7)$ and $\mathbf{d_2} = (-2, 6, 7)$. What is the relationship between the two lines (assuming they are coplanar)?



a. perpendicular

b. parallel

c. identical

d. neither parallel nor perpendicular

44 [Challenging] • Target: The Scalar (Dot) Product & Angles • Cross-Lesson: 2.01 Exponentials & Logarithms

Two lines have direction vectors $\mathbf{d_1} = (-3, 6, -2)$ and $\mathbf{d_2} = (2, 1, 7)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. neither parallel nor perpendicular

c. parallel

d. perpendicular

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45 [Challenging] • Target: Vector Equations of Lines in 3D • Cross-Lesson: 3.01 Advanced Trigonometry

Two lines have direction vectors $\mathbf{d_1} = (4, 5, 8)$ and $\mathbf{d_2} = (-4, 7, -1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. perpendicular

c. parallel

d. neither parallel nor perpendicular

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